

Week 01: Statistics and Mathematics for ML

Module 01: Descriptive Statistics and Distributions

- Mean, median, mode: when each is appropriate
- Variance and standard deviation as measures of spread
- Percentiles, quartiles, IQR, and z-score
- Distribution shapes: symmetric vs skewed, long tails; outlier detection with IQR fences
- When median and IQR are preferred over mean and standard deviation

Module 02: Probability Basics for ML

Module 2.5: Worked examples on center, spread, IQR fences, and z-scores

Module 03: Data Quality, Scaling, and Encoding

Module 3.5: Bayes and confusion-rate worksheets

Module 4: Quiz

What is Statistics & Statistics for ML

What is Statistics?

Statistics is the science of working with data — **collecting, summarizing, analyzing, and interpreting** it to understand the real world through numbers.

Example:

Suppose we collect the marks of all students in a class.
The **mean** tells us the class's average performance,
the **median** shows the middle score,
and the **standard deviation** tells us how spread out the marks are.
That's statistics — finding meaning in data.

What is Statistics for Machine Learning?

Statistics for ML helps us **understand the data before we train models**.

A model learns patterns from data — so if we don't understand our data's behavior, we risk building biased or misleading models.

We use statistics in ML to study **the center, spread, shape, and relationships** within data.

It forms the foundation for preprocessing, feature scaling, and model evaluation.

Types of Statistics

There are two main types of statistics:

1. **Descriptive Statistics** — describe or summarize data.

Example: the average mark of a class is 78.

2. **Inferential Statistics** — draw conclusions about a population from a sample.

Example: using one class's marks to estimate the average for the whole university.

In machine learning, we mostly rely on **descriptive statistics** because we focus on **understanding and preparing our dataset**, not on generalizing to an unseen population."

Module 01: Descriptive Statistics and Distributions

Module 01 Overview

Descriptive statistics helps us **summarize and visualize** data using numbers and graphs. We'll learn how to describe data's center, spread, and shape.

Topics Covered:

- Mean, Median, Mode — and when each is appropriate
- Variance and Standard Deviation — measuring spread
- Percentiles, Quartiles, IQR, and Z-Score
- Distribution Shapes — symmetric vs skewed, outliers via IQR fences
- When Median and IQR are preferred over Mean and SD

Example:

In salary data, if one CEO earns far more than others, the mean shoots up. The **median**, however, better represents a “typical” salary. That's why median and IQR are often safer for skewed data.

Module 02: Probability Basics for ML

Module 02 – Probability Basics

Next, we'll study **probability**, the mathematics of uncertainty.

We'll discuss events, sample space, conditional probability, and independence.

We'll intuitively understand **Bayes' Theorem**, the core idea behind Naïve Bayes classification.

Then we'll explore **sensitivity, specificity, false positive and negative rates**, and how **class imbalance** can distort evaluation metrics — critical knowledge for healthcare, fraud, or sentiment-analysis models.

Module 2.5 and 3 Overview

From Theory to Practice

In Module 2.5, we'll do worked examples — computing center, spread, IQR fences, and Z-Scores by hand and in Python.

Then Module 03 takes us into **Data Quality, Scaling, and Encoding** — the practical bridge between statistics and machine learning.

We'll cover:

- Types of missing data (MCAR, MAR, MNAR) and simple imputation
- Standardization, Min-Max, and Robust Scaling — formulas and use cases
- Nominal vs Ordinal Encoding and their geometric impact
- Distance metrics (Euclidean, Manhattan, Cosine)
- Covariance, Correlation, and a conceptual intro to PCA

Module 3.5 and 4 Overview

Finally, in **Module 3.5**, we'll apply what we've learned through **Bayes' and confusion-rate worksheets** — completely hands-on, no coding required.

Then, **Module 4** wraps up this unit with the **Concept Quiz**

These final modules help connect everything, understanding data, applying probability, and evaluating model performance

before we move into coding in later weeks.

Week's Goal

This Week's Goal

Week 1 is about building intuition.

You'll learn how to summarize data, detect outliers, understand distributions, and connect these concepts to machine learning tasks.

Remember: We're not training to be statisticians, but to use statistics as a tool to make machine learning models accurate, robust, and explainable.